

CHURN.AI Workflow & Parameters Documentation

1. AI Prediction API Workflow

The primary workflow is the predictive API that calculates the risk of churn using logistic regression models for 5 business domains: Telecom, E-Commerce, Banking, SaaS, and Gaming.

When a user accesses the platform, they utilize this workflow to get instantaneous churn risk scores.

Process Flow:

1. User Logs In.
2. Selects Domain (e.g., SaaS, Telecom).
3. Enters Parameters (up to 13 parameters depending on domains like tenure, TotalCharges, npsScore, etc.).
4. The React Client posts to `/api/predict`.
5. The API scores using a Logistic Regression formula:

$$P(\text{churn}) = \text{sigmoid}(b_0 + b_1x_1 + \dots + b_nx_n)$$

6. Output: Score (0-100), Risk Tier (LOW, MODERATE, HIGH, CRITICAL), and factor explainability breakdown.

Example Parameters (SaaS domain):

- subscriptionAge, planTier, monthlyActiveDays, featuresUsedPct, teamSize, lastLoginDaysAgo, apiCallsMonthly, supportTickets, npsScore, hasIntegrations, hasSso, hasDedicatedCsm.

2. ML Model Training Pipeline Workflow

The secondary workflow is the internal ML engine for regenerating churn models, implemented in `ml\_engine/train\_xgboost\_churn.py`.

Process Flow:

1. Data Ingestion: Connects to the data warehouse and extracts user features (e.g., tenure, monthly_charges, total_charges, support_tickets_6mo, nps_score, active_days_last_month, usage_drop_pct).
2. Label Generation: Uses simulated churn logic based on usage drops and NPS.
3. Data Splitting: Splits into 80% training data and 20% test data.

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4. Normalization: Uses StandardScaler to scale continuous features.
5. Hyperparameter Definition & Optimization: Initialize XGBoost.
6. Execution: GPU cluster convergence training.
7. Evaluation metrics generated: Accuracy, Precision, Recall, F1-Score.
8. Artifact Export: Saves the trained model as 'xgboost_churn_v2.pkl'.

Key Hyperparameters for XGBClassifier:

- n_estimators = 200
- learning_rate = 0.05
- max_depth = 6
- subsample = 0.8
- colsample_bytree = 0.8
- eval_metric = logloss

These two workflows represent the entire lifecycle of the Churn.AI platform, balancing realtime logistical regression models with heavier off-line XGBoost models.